

WEEK 2

STATISTICAL INFERENCE REVIEW

APPLIED STATISTICAL ANALYSIS/QUANTITATIVE METHODS I

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ROAD MAP FOR TODAY

- Lecture recap
- Exercises

KEY CONCEPTS AND TERMS

- **Unit of analysis:** The observation described by a set of data

- ▶ Ex: Voters, parties, bills, elections, and voting decisions

- **Variable:** A characteristic or attribute of the unit of analysis that can vary across observations

“A characteristic that can vary in value among subjects in a sample or population.” (Agresti and Finlay 2009, 11)

- **Outcome variable:**

- ▶ Also called response or *dependent* variable
- ▶ Denoted as Y
- ▶ Phenomenon we aim to explain

KEY CONCEPTS AND TERMS

■ Predictor variable:

- ▶ Also called the *input variable* or *covariate*
- ▶ Denoted as X
- ▶ Explains variation in outcome

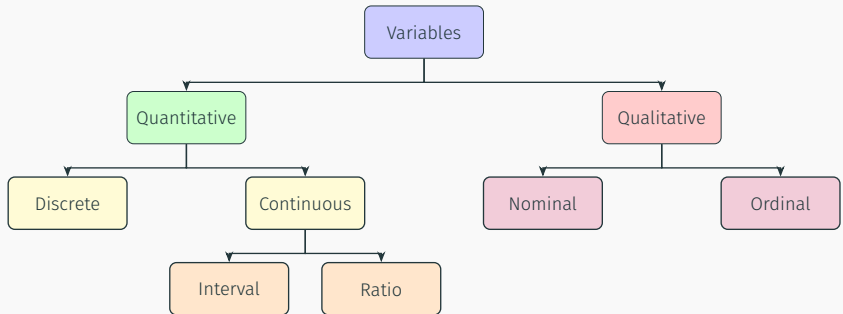
■ What is variation? Ex: Age $\rightarrow$ Income

MEASUREMENT

Definition

Measurement refers to the way variables are quantified

Ex: Economic wealth may be measured as GDP



MEASUREMENT: CLASSIFYING VARIABLES

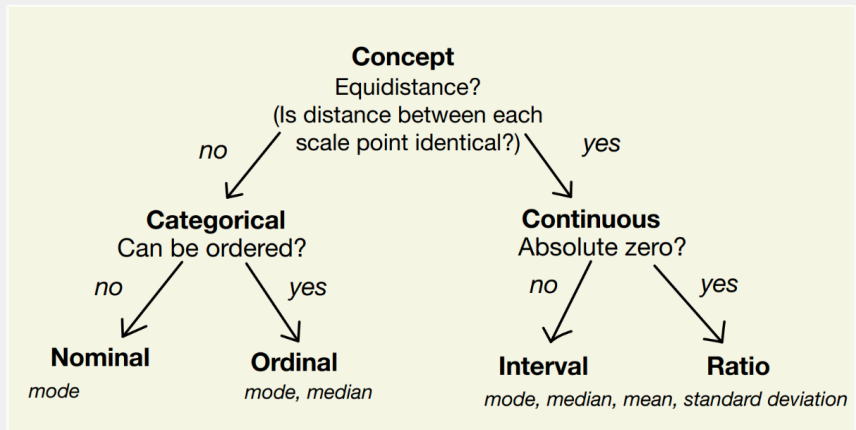
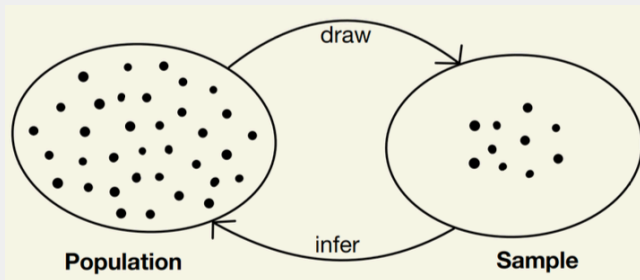


Figure: Adapted from Kellstedt and Whitten (2018), Chapter 5

POPULATION AND SAMPLE

- What is relationship between population and sample?
 - ▶ **Population:** Total set of subjects of interest in a study
 - ▶ **Sample:** Subset of population on which the study collects data.
 - ▶ **Parameter:** Numerical summary of population
 - ▶ **Statistic:** Numerical summary of sample data



DESCRIPTIVE AND INFERENCE STATISTICS

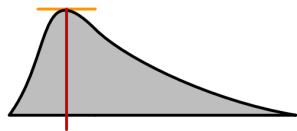
- **Descriptive statistics:** “Summarize the information in a collection of data.” Agresti and Finlay (2009, 4).
- **Inferential statistics:** “Provide predictions about a population, based on data from a sample of that population.” Agresti and Finlay (2009, 4).

DESCRIPTIVE STATISTICS

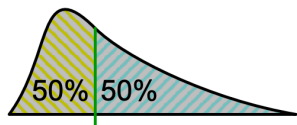
Measures of Central Tendency and Variability

- **Central tendency:** Mean, median, mode
- **Variability:** Variance, standard deviation, range, IQR
- **Visualization:** Boxplots

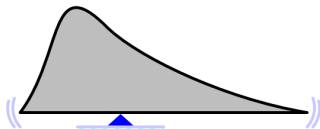
DESCRIPTIVE STATISTICS: CENTRAL TENDENCY



mode



median



mean

- **Mode:** The most frequently occurring value of X
 - ▶ Some distributions can have more than one mode
- **Median:** Value of X that falls in the middle position when observations are ordered from smallest to largest
 - ▶ Median = 50th percentile
= second quartile
- **Mean:** Most common measure of central tendency

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

COMPARING MEASURES OF CENTRAL TENDENCY

- In a perfectly symmetric distribution, such as a normal distribution:

$$\text{mode} = \text{median} = \text{mean}$$

- Not true when distribution is non-symmetric

- ▶ Right-skewed distribution (positive skew):

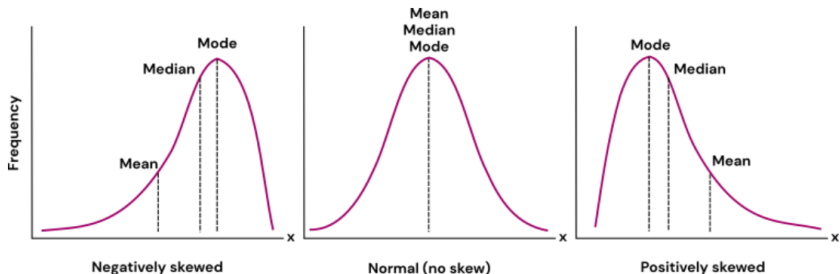
$$\text{median} < \text{mean}$$

- ▶ Left-skewed distribution (negative skew):

$$\text{median} > \text{mean}$$

- Mean is sensitive to outliers, while median is more robust

COMPARING MEASURES OF CENTRAL TENDENCY



DESCRIPTIVE STATISTICS: VARIABILITY

- Sample variance: Average of squared deviations from mean

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

- Why do we average by dividing by $n - 1$ rather than n ?
- Because deviations from sample mean must sum to zero
- Once $n - 1$ deviations are known, last deviation is fixed
- Thus, only $n - 1$ squared deviations vary freely; these are *degrees of freedom*
- Sample standard deviation: Square root of sample variance

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

DESCRIPTIVE STATISTICS: VARIABILITY

- **Range:** Difference between largest and smallest measurement

$$\text{Range} = x_{\max} - x_{\min}$$

- **Interquartile range (IQR):** Difference between upper and lower quartiles; range of middle 50% of distribution

$$\text{IQR} = x_{Q3} - x_{Q1}$$

PROBABILITY

- What is a probability?
- What is a distribution?
- What is a probability distribution?

PROBABILITY: BASIC TERMINOLOGY

- **Experiment:** repeatable procedure for making an observation
- **Outcome:** Possible result of such an experiment
- **Sample space:** (Ω) of an experiment is the set of all possible outcomes
- **Event:** Subset of the sample space, that is, any set of outcomes
- **Probability** of an event is its long-run relative frequency

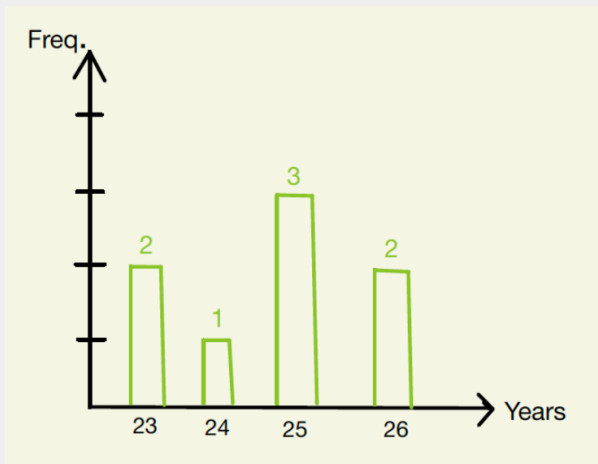
PROBABILITY: BASIC TERMINOLOGY

- If $\Pr(A) = 0.5$, then event A occurs approximately half of time when experiment is repeated infinitely often
- When an experiment is repeated many finite times, observed relative frequency is expected to become closer to probability as number of repetitions increases

$$P(A) = \frac{\text{Number of elements in } A}{\text{Number of all elements}}$$

PROBABILITY: DISTRIBUTION

Ex: Age of people in the room



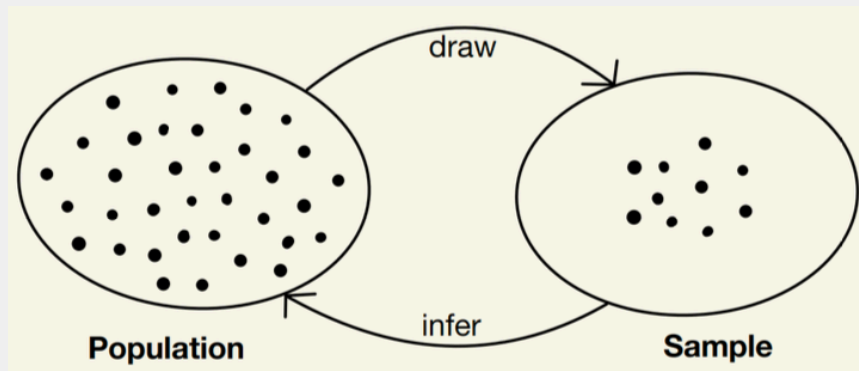
A distribution can take different shapes and therefore has different names, such as normal, binomial, or t-distribution

PROBABILITY: PROBABILITY DISTRIBUTION

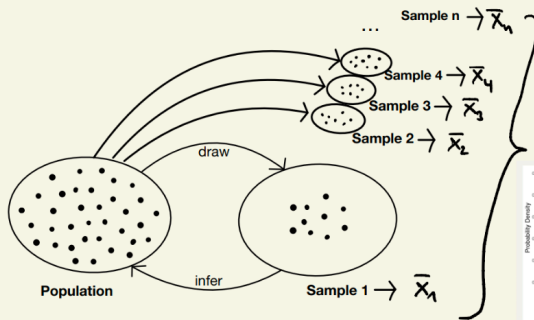
- Distributions of random variables are **probability distributions** if, for all possible outcomes, they specify probabilities that those outcomes occur
- Probability distributions are analogous to **frequency distributions**, except that they are based on probability theory rather than observations in sample data
- Agresti and Finlay (2009, 75): “lists the possible outcomes and their probabilities”

SAMPLING DISTRIBUTION

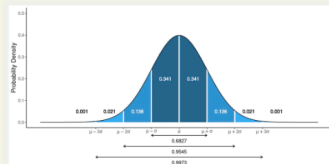
Recall the basic idea for empirical research:



SAMPLING DISTRIBUTION



Sampling distribution



SAMPLING DISTRIBUTION

- “A sampling distribution of a statistic is the probability distribution that specifies probabilities for the possible values the statistic can take” Agresti and Finlay (2009, 87)
- Suppose a population has mean μ and variance σ^2 , and we are interested in estimating the population mean
- Repeatedly taking samples from that population and calculating mean for each sample yields **sampling distribution of mean**

SAMPLING DISTRIBUTION: WHY IS THIS IMPORTANT?

- Probability theory helps us predict how close a statistic falls to parameter it estimates

How close is $\bar{x}$ to μ ?

- A point estimate is a single number that is best guess for parameter value
- If we repeatedly take samples, mean of sample means equals population mean μ in long run
- **Standard error** describes how much $\bar{x}$ varies from sample to sample

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

CENTRAL LIMIT THEOREM

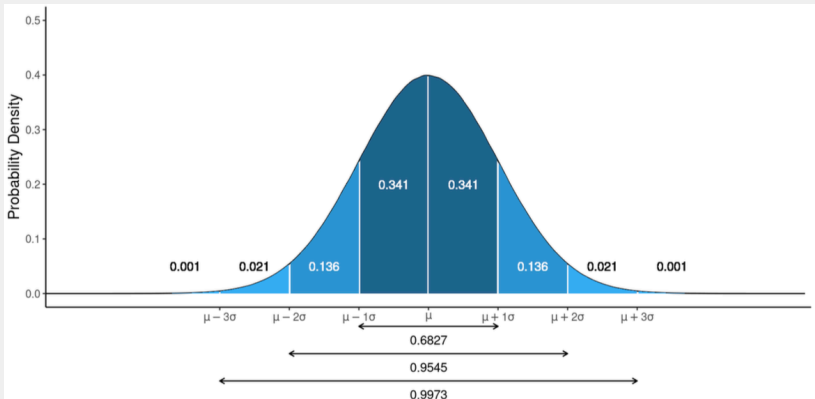
Statement

Sampling distribution of a statistic approaches a normal distribution with mean μ and variance σ^2/n as # of samples n increases

- This holds regardless of shape of original population distribution
- Take a sample, calculate its mean, and repeat the procedure many times
- Resulting distribution of sample means approaches a normal, bell-shaped curve, *even if population distribution was not normal*
- CLT provides a basis for applying statistical inference to many natural and social phenomena (and almost everything we will do in this class!)

NORMAL DISTRIBUTION

- A continuous distribution that describes data clustered around mean
- It is uniquely determined by its mean, median, and mode μ , as well as its variance σ^2



CONFIDENCE INTERVALS

Definition

A confidence interval is an interval of numbers around a point estimate that we believe contains population parameter

Confidence interval = Point estimate $\pm$ Margin of error

CONFIDENCE INTERVALS

- Our estimate of a population parameter varies across repeated samples, generating a sampling distribution
- Instead of relying only on a point estimate, we can use an **interval estimate**: a range within which true parameter lies with a specified "level of confidence"
- CIs can be constructed using standard error or variance of an estimate
- A $q\%$ CI likely contains true parameter in at least $q\%$ of repeated samples
- Interactive visualization:
<https://rpsychologist.com/d3/ci/>
- **Important:** A $q\%$ CI does *not* mean that there is a $q\%$ probability that fixed population parameter lies inside this particular interval!

CLASS BUSINESS

- Read required (and suggested) online materials
- Problem set # 1 is up on GitHub